

Assessment Test: Software Tester (QA)

Candidate Response - Task Management Application

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1.Approach and Assumptions

No test documentation or working build was provided, so the test cases below were written directly from the feature list given in the assessment. Standard black-box techniques were used - equivalence partitioning, boundary value analysis, and error guessing - and each functional area is broken into scenarios with positive, negative, and edge cases.

Since the application spec was not detailed, the following assumptions were made:

- Registration requires Name, Email, Password, and Confirm Password.
- Login requires Email and Password.
- A task has a mandatory Title, and optional Description / Due Date.
- The app follows a typical web architecture with a backend database and session-based login. •Each user can only access their own tasks (basic multi-user data isolation).

1. Test Scenarios and Test Cases

2.1 Registration

Scenario: Register with valid and invalid details

TC ID	Title	Steps	Expected Result	Type
REG-01	Register with all valid, unique details	1. Open Registration page 2. Enter valid name, unique email, matching password/confirm password 3. Click Register	Account created; success message shown; user redirected to Login or dashboard	Positive
REG-02	Register with minimum valid password length	Enter password exactly at minimum allowed length (e.g. 8 characters)	Registration succeeds	Edge
REG-03	Register with already registered email	Use an email already present in the system	Registration blocked; error - "Email already registered"; no duplicate account created	Negative
REG-04	Register with invalid email format	Enter email without "@" or domain, e.g. user.example.com	Validation error shown; form not submitted	Negative
REG-05	Register with mismatched password and confirm password	Enter different values in Password and Confirm Password	Error - "Passwords do not match"; registration blocked	Negative

REG-06	Register with empty mandatory fields	Leave Name / Email / Password blank and submit	Field-level validation errors shown for each blank mandatory field	Negative
REG-07	Register with weak password (below policy)	Enter password shorter than minimum, or without required complexity	Error indicating password policy not met; registration blocked	Negative
REG-08	Register with maximumlength inputs	Enter name/email at the exact maximum allowed length	Data accepted and stored correctly, no truncation	Edge
REG-09	Register with input exceeding max length	Enter name/email one character beyond max allowed length	Input rejected/truncated with a clear message; no server error	Edge
REG-10	Register with special characters / script in name field	Enter values such as <script>alert(1)</script> or ' OR '1'='1 in Name	Input sanitised/escaped; no script execution or DB error; stored as plain text	Edge
REG-11	Register with leading/trailing spaces in email	Enter " user@example.com " with surrounding spaces	Spaces trimmed automatically, or clear validation error - behaviour should be consistent	Edge
REG-12	Register with case-variant duplicate email	Register "User@Example.com" when "user@example.com" already exists	Treated as duplicate (email should be caseinsensitive)	Edge

2.2 Login

Scenario: Authenticate a registered user

TC ID	Title	Steps	Expected Result	Type
LOG-01	Login with valid registered credentials	Enter correct email and password; click Login	User authenticated; redirected to task list/dashboard; session created	Positive
LOG-02	Login with incorrect password	Enter valid email with wrong password	Generic error - "Invalid email or password" (should not reveal whether the email exists)	Negative
LOG-03	Login with unregistered email	Enter an email not present in the system	Same generic error as LOG-02, to prevent user enumeration	Negative
LOG-04	Login with empty email/password fields	Leave one or both fields blank and submit	Validation error shown; request not sent with empty credentials	Negative
LOG-05	Login with correct email, wrong-case password	Enter password with different letter casing than registered	Login fails (password must be case-sensitive)	Edge
LOG-06	Login with email in different case	Enter "USER@Example.com" instead of registered "user@example.com"	Login succeeds; consistent with caseinsensitive email handling at registration	Edge
LOG-07	Repeated failed login attempts	Attempt login with wrong password 5-10 times in a row	Account temporarily locked, or CAPTCHA/ratelimit triggered	Edge
LOG-08	Login with SQL injection style input	Enter ' OR '1'='1 in email/password fields	Login rejected; input treated as literal text; no unauthorised access or DB error	Negative
LOG-09	Session persists after login	Login successfully, refresh the page	User remains logged in; session/token persists as expected	Positive
LOG-10	Access task pages directly without logging in	Without logging in, navigate directly to the task list URL	Redirected to Login page; task data not exposed	Negative
LOG-11	Logout invalidates the session	Login, click Logout, then press browser Back	Protected pages are not accessible via cached/back navigation; re-login required	Edge

2.3 Task CRUD Operations

Scenario: Create Task

TC ID	Title	Steps	Expected Result	Type
TC-01	Create task with valid title	Login → Create Task → enter valid title → Save	Task created and immediately visible in task list	Positive
TC-02	Create task with all fields filled	Enter title, description, and due date; Save	Task saved with all fields correctly persisted and displayed	Positive
TC-03	Create task with empty title	Leave Title blank; click Save	Validation error - "Title is required"; task not created	Negative
TC-04	Create task with only whitespace as title	Enter only spaces in the Title field; Save	Treated as empty; validation error shown, task not saved blank	Edge
TC-05	Create task with maximum allowed title length	Enter title at the exact max character limit	Task saved successfully, no truncation	Edge
TC-06	Create task with title exceeding max length	Enter title beyond the allowed max length	Input rejected or truncated with a clear message; no server error	Edge
TC-07	Create task with special characters/emoji in title	Enter a title like "Fix bug #123 @urgent!"	Task saved and displayed correctly, characters intact	Edge
TC-08	Create task with a past due date	Set due date to before today; Save	App either warns the user or accepts and flags the task as overdue; behaviour should be consistent	Edge
TC-09	Create duplicate task (same title twice)	Create two tasks with identical titles	Both are created independently unless a business rule states otherwise	Edge
TC-10	Create task during a network failure	Disconnect network mid-submit and click Save	User-friendly error shown; task not partially saved; retry possible	Negative

Scenario: View Task List

TC ID	Title	Steps	Expected Result	Type
TC-11	View task list with existing tasks	Login as a user with existing tasks; open task list	All of that user's tasks display accurately	Positive
TC-12	View task list with zero tasks	Login as a new user with no tasks	Empty-state message shown (e.g. "No tasks yet"), not a blank screen	Edge
TC-13	View task list with a large number of tasks	Login as a user with 500+ tasks	List loads within acceptable time; pagination/lazy loading works; no freeze	Edge
TC-14	Verify data isolation between users	Login as User A, note tasks; logout; login as User B	User B sees only their own tasks, none of User A's	Negative
TC-15	Task list reflects changes without reload	Create/edit/delete a task and return to the list	List updates to reflect the latest state accurately	Positive

Scenario: Edit Task

TC ID	Title	Steps	Expected Result	Type
TC-16	Edit task with valid new data	Open an existing task → change title/description → Save	Changes persisted and reflected immediately	Positive
TC-17	Edit task and clear the mandatory title	Open task, delete all text in Title, click Save	Validation error shown; original data preserved until valid input given	Negative
TC-18	Edit a task that was just deleted elsewhere	Open task for edit in one tab; delete the same task in another tab; try to Save	Graceful error - "Task no longer exists"; no crash or ghost record	Edge
TC-19	Concurrent edit from two sessions	Open the same task in two tabs; edit and save both with different values	Defined conflict behaviour (last write wins, or a conflict warning) - no silent data loss	Edge

TC-20	Edit another user's task via ID manipulation	Login as User A; change task ID in the URL/request to one owned by User B; attempt edit	Request denied (403/Forbidden); unauthorised edit blocked	Negative
TC-21	Cancel edit without saving	Open task, change fields, click Cancel	Original task data remains unchanged	Positive

Scenario: Delete Task

TC ID	Title	Steps	Expected Result	Type
TC-22	Delete an existing task	Select a task → Delete → confirm	Task removed from the database and the list	Positive
TC-23	Cancel a delete confirmation	Click Delete, then Cancel on the confirmation dialog	Task is not deleted; remains in the list unchanged	Positive
TC-24	Delete an already-deleted task (double request)	Trigger delete twice quickly (double-click or replayed request)	Second request handled gracefully; no error or crash	Edge
TC-25	Delete another user's task via ID manipulation	Login as User A; attempt to delete a task ID belonging to User B	Request denied (403/Forbidden); User B's task remains intact	Negative
TC-26	Delete task and verify list/count updates	Note total task count; delete one task; check the list	Task count decreases by exactly one; UI refreshes correctly	Positive

2.4 Input Validation and Error Handling

Scenario: General validation across all forms

TC ID	Title	Steps	Expected Result	Type
VAL-01	Submit a form with all valid data	Fill any form (Registration/Task) correctly and submit	Form submits successfully with no falsepositive errors	Positive
VAL-02	Bypass client-side validation via direct API call	Send an invalid payload (e.g. empty title) directly to the backend, bypassing the UI	Server-side validation independently rejects the invalid data	Negative
VAL-03	Submit HTML/script tags in a text field	Enter in task title/description	Input is escaped on output; no stored XSS executes when the list renders	Negative
VAL-04	Submit an oversized payload	Enter a very large amount of text into a description field	Server rejects with a clear error or enforces max length; no crash/timeout	Edge
VAL-05	Submit non-numeric data in a numeric field	If a field expects a number, enter letters instead	Validation error; type mismatch handled gracefully	Negative
VAL-06	Submit an invalid date	Enter something like 31/02/2026 in the due date field	Validation error shown; invalid date rejected	Negative
VAL-07	Server error during submission	Simulate the backend/database being unavailable during submit	User sees a friendly error message, not a raw stack trace; no data loss client-side	Negative
VAL-08	Rapid repeated submission (double-submit)	Click Submit multiple times quickly	Only one record is created; button disabled after first click, or request de-duplicated	Edge
VAL-09	Unicode / multilingual input	Enter non-Latin text (Hindi, Chinese, emoji) in any text field	Input accepted, stored, and displayed correctly without corruption	Edge
VAL-10	Leading/trailing whitespace on paste	Paste " Buy groceries " into the task title	Whitespace trimmed before save, or consistently preserved	Edge

2. Bug Identification / Risk Areas

The following were identified without executing the application, based on the feature description and common failure patterns in CRUD-based web apps with authentication. These are the risk areas the test cases above are designed to catch.

#	Potential Bug / Risk Area	Severity	Reason / Impact
1	Missing authorisation check could allow a logged-in user to view/edit/delete another user's tasks by changing the task ID in the URL or request (Insecure Direct Object Reference)	Critical	Direct breach of data privacy between users; could expose or corrupt another user's data
2	No server-side validation - the app may rely only on client-side checks for required fields, lengths, and formats	Critical	Client checks can be bypassed via browser tools or direct API calls, letting invalid or malicious data into the database
3	No protection against SQL injection or stored XSS in task title/description or registration fields	Critical	Could allow database compromise or malicious scripts executing in other users' browsers when the task list renders
4	No account lockout or rate-limiting on repeated failed login attempts	Major	Application is vulnerable to brute-force password guessing
5	Session may not be invalidated properly on logout (Back button could still show protected pages from cache)	Major	Security risk on shared/public computers; unauthorised access to a previous session's data
6	No pagination or lazy loading for the task list when a user has a large number of tasks	Major	Performance degradation and slow loads as data grows - a scalability concern close to production
7	No delete confirmation, or deletion is not reversible (no undo/soft-delete)	Major	Risk of accidental, permanent data loss for users
8	Duplicate email check may only happen in application logic, not at the database level (possible race condition)	Major	Two simultaneous registrations with the same email could both succeed, creating conflicting accounts
9	Inconsistent or missing whitespace trimming on text inputs (email, task title)	Minor	Leads to confusing duplicate-looking entries and inconsistent search/filter results
10	Raw/generic error messages (e.g. server stack traces) may be exposed to the user on backend errors	Minor	Poor user experience and possible information disclosure about backend architecture
